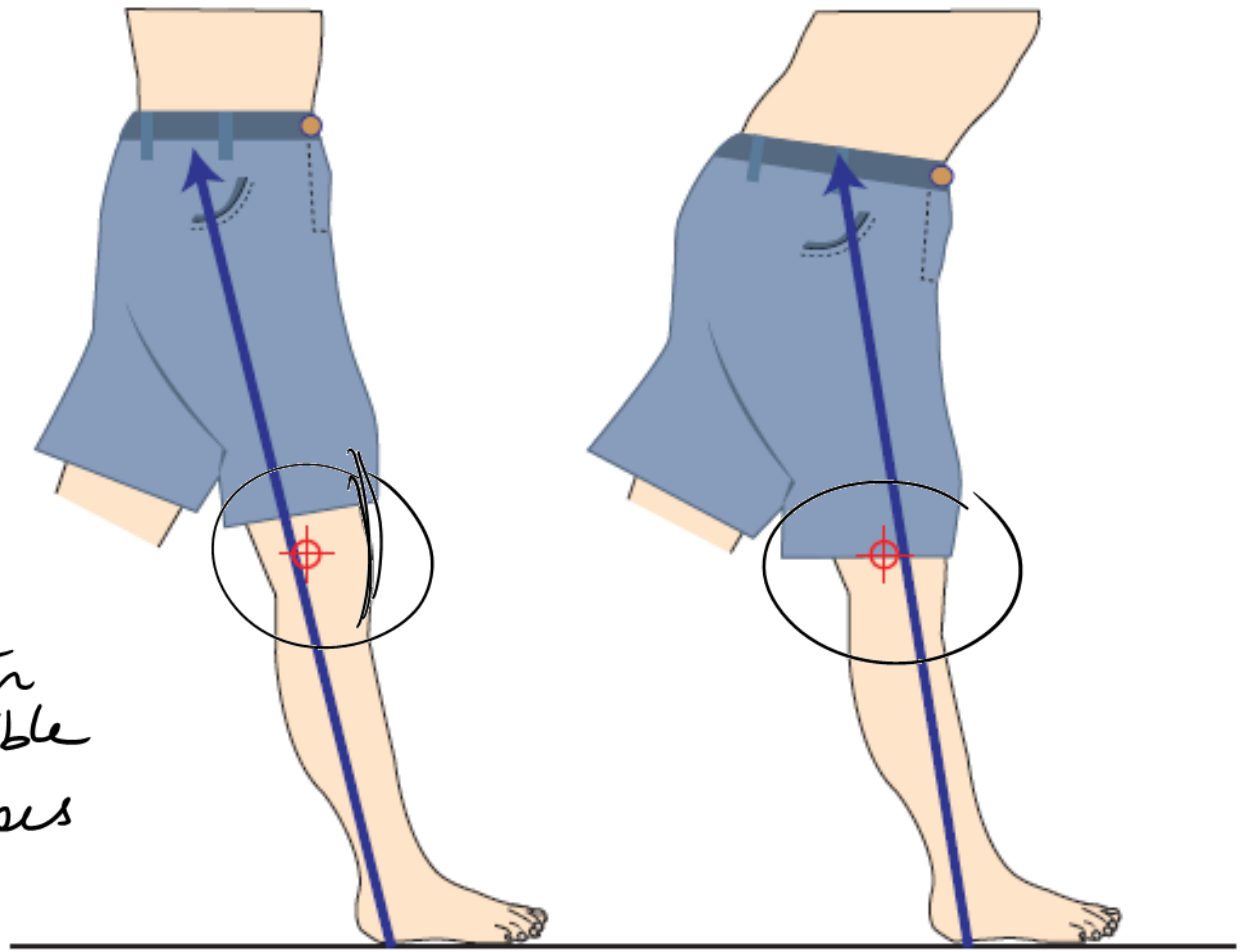
Pathological gait – Anterior Trunk Bending

Purpose may be to compensate for weak knee extensors

Person may also place one hand on the thigh, on the affected side to stabilize the knee

Weak hip extensors
Hip flexion contracture

} other possible causes



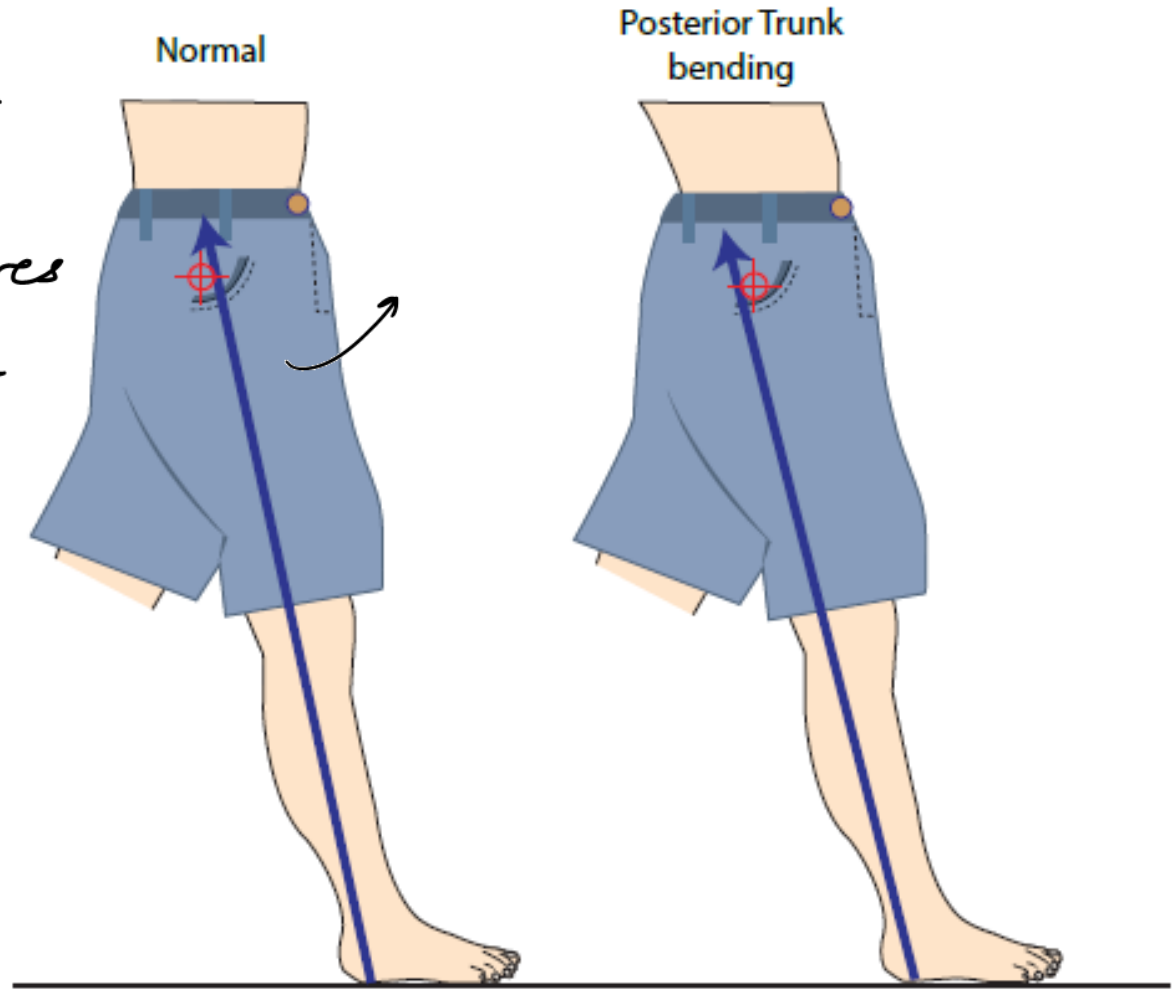
Pathological gait – Posterior Trunk Bending

Observed in the sagittal plane

Reversed version of anterior trunk bending

Early in stance, whole trunk moves backwards in the sagittal plane

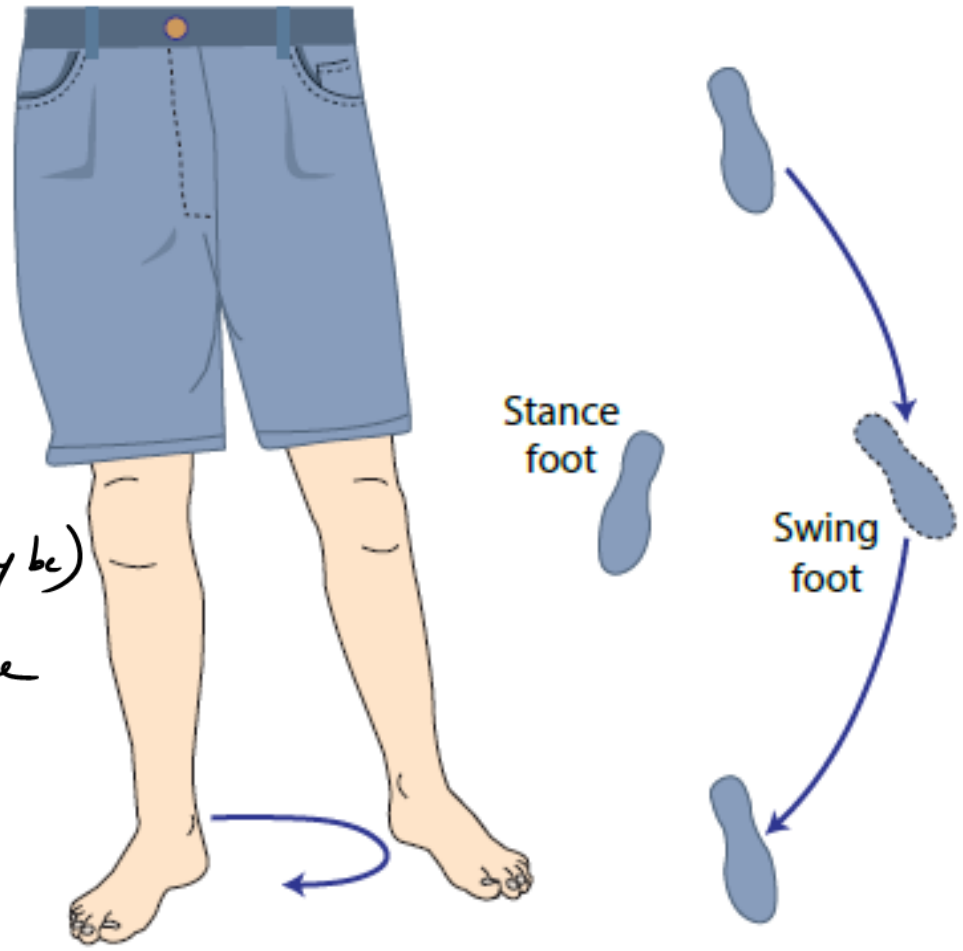
Compensate for weak hip extensors.



Functional leg length discrepancy

Anatomical leg length discrepancy
– legs are actually different lengths

"Functional" leg length discrepancy
– legs are not necessarily of different lengths (although they may be) but one or both legs are unable to attain the appropriate length for a particular phase of the gait cycle.



For normal walking
Stance phase leg needs to be longer than the swing leg. If it is not, the swinging legs collides with the ground and is unable to move forward past the stance leg.

Legs functionally lengthened in the stance phase by extending the hip & the knee and plantarflexing the ankle

Leg functionally shortened for the swing phase by flexing the hip & the knee and dorsiflexing the ankle

Spasticity (tightness) of extensors or weakness of flexors cause problems in the swing phase

Also mechanical locking of knee / hip joint in extension

Conversely, spasticity of flexors or weakness of extensors cause problems in limb lengthening in the stance phase.

→ Increase in functional leg length - longer than necessary in swing

Following a 'stroke' - paralysis in the tibialis anterior - causes foot drop

Increase in tone in the hip & knee extensor muscles

Gait modifications — either lengthen the stance leg or shorten the swing leg or use a combination.

Functional leg length discrepancy

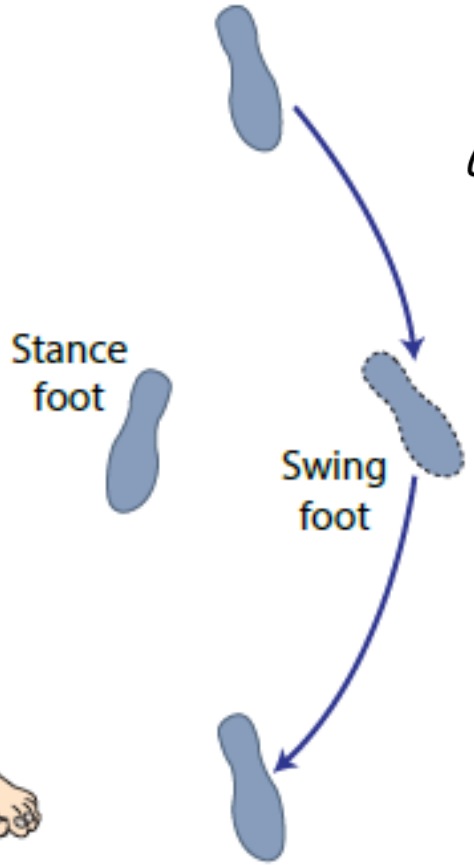
Anatomical leg length discrepancy
– legs are actually different lengths

Circumduction: avoid ground contact of the swinging leg by swinging it outward



1 Circumduction

flexion/extn
&
abduction/adduction

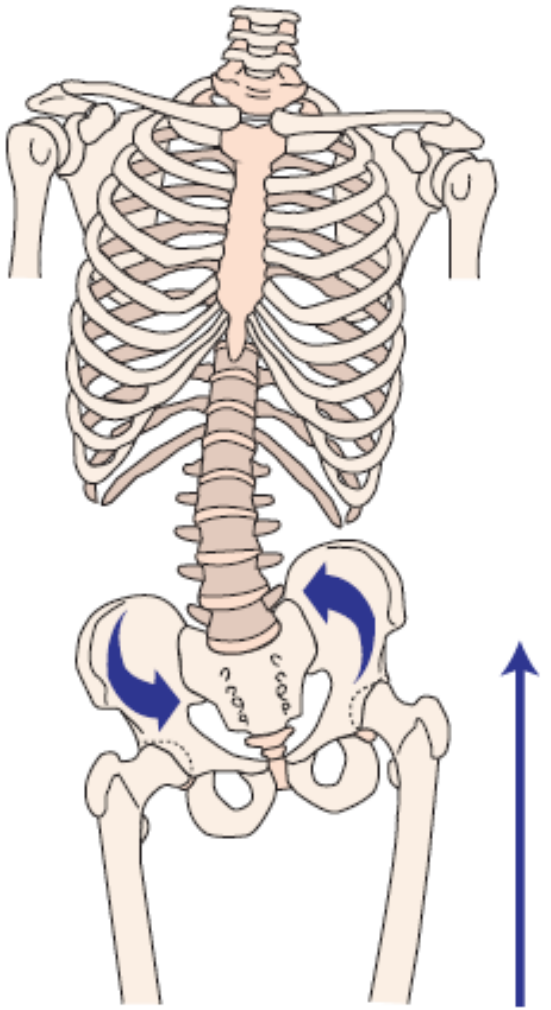


Functional leg length discrepancy

(2) Hip hiking

Pelvis is lifted on the side of the swinging leg

Commonly used in slow walking with weak hamstrings, because knee will extend prematurely and therefore the leg will be too long towards the end of swing



Functional leg length discrepancy

3 Steppage

Swing phase modification

Use exaggerated hip & knee flexion to lift the foot higher than usual

Used to compensate for a plantarflexed ankle (foot drop) due to inadequate dorsiflexor control



Functional leg length discrepancy

(4) Vaulting

The person goes up on the toes of the stance leg to enable ground clearance of the swing leg

Stance phase modification

The first 3 are swing phase modifications

